Zehua Wang

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Education

- **Massachusetts Institute of Technology — GPA: 5.00/5.00** Cambridge, MA
  B.S. Candidate in Physics and Artificial Intelligence and Decision Making *Expected May 2028*
- **Tsinghua University — GPA: 3.95/4.00** Beijing, China
  Preparatory Program and First-Year Coursework, Institute for Interdisciplinary Information Sciences (IIIS) *Aug 2024 – Jul 2025*

Honors & Awards

- *Gold Medal* ***(1st in Theory)***, 54th International Physics Olympiad (IPhO) *Jul 2024*
- *1st Place*, MIT Informatics Tournament (MITIT) 2025 Winter Contest Beginner's Round *Dec 2025*
- *2nd Place, Modal Prize*, HackMIT 2025 *Sep 2025*

Experience

- **Sophomore Intern, Algorithm Development & Software Engineering** New York, NY
  Hudson River Trading *May 2026 – Aug 2026*
  - Developed and live-tested a trading strategy during the Algorithm Development rotation.
  - Contributed enhancements to internal developer tooling in HRT's codebase during the Software Engineering rotation.
- **Undergraduate Researcher, Diffusion Models & Molecular Dynamics** Cambridge, MA
  MIT — Advisor: Tommi Jaakkola *Feb 2026 – May 2026*
  - Co-developed Diffusion Accelerants, a non-Markovian biasing method that augments molecular dynamics with learned measure transport.
- **Undergraduate Researcher, Reinforcement Learning & Embodied AI** Cambridge, MA
  FortyFive Labs — Advisor: Ge Yang *Oct 2025 – Apr 2026*
  - Developed a whole-body control policy for functional compliance, combining explicit force estimation with directional, time-varying stiffness profiles at deployment.
  - Ran parallel RL and imitation-learning experiments across nearly 100 NVIDIA RTX PRO 6000 GPUs; performed extensive sim-to-real deployment, debugging, and policy iteration on Unitree G1 hardware.
  - Contributed code to **Vuer**, a 3D visualization toolkit, and **ML-Dash**, an experiment-tracking and data platform.
- **Undergraduate Researcher, Learning-Based Control** Beijing, China
  Tsinghua University — Advisor: Huazhe Xu *Mar 2025 – Aug 2025*
  - Developed a multistage RIR (RL-to-imitation-to-real-world) pipeline for Franka manipulation, combining PPO/DrQ-v2 specialists, multitask imitation learning, and sim-to-real transfer.

Publications

- **Diffusion Accelerants: Towards Augmenting Molecular Dynamics with Learned Measure Transport**
  Bowen Jing, **Zehua Wang**, Paul Gutkovich, Peter Holderrieth, Tommi Jaakkola.
  SPIGM Workshop at ICML 2026 (poster). [Paper]
- **Family-Aware Residual Architecture for Predicting Quantum Circuit Simulation Performance**
  Honjar Xing*, Yehong Jiang*, Xianbang Wang*, **Zehua Wang***, Zhicheng Jiang*.
  QC-CSAA Workshop at IEEE ISVLSI 2026 (full paper). [Paper] * Equal contribution.

Selected Projects

- **Open-Source Contributor, NousResearch/Hermes Agent**: Authored a **3-commit** speech-to-text provider-validation patch now in `main`, adding selected-provider and plugin-backend checks plus tests. [Repo] [Commits] [Review]
- **Video Real2Sim (VR2S) (MIT 6.8300)**: Co-built a single-image real-to-sim pipeline that synthesizes orbital views with a video model and reconstructs them with a pose-free 3D stack; improved perceptual fidelity and back-side appearance while characterizing a silhouette-accuracy tradeoff. [Project]
- **Fast Humanoid Loco-Manipulation via Flow Matching (MIT 6.4210)**: Compared DDPM and Flow Matching for humanoid trajectory synthesis; at five function evaluations, Flow Matching achieved **820 vs. 280 survival steps**. Enabled zero-shot loco-manipulation from walking-only data via test-time classifier guidance. [Code] [Video]
- **RL vs. SFT for Mathematical Reasoning in LLMs (MIT 6.4610)**: Implemented GMPO and its evaluation pipeline for a compute-controlled comparison of PPO, GRPO, GMPO, and RLOO with SFT on Qwen3-8B; GMPO reached **74.2%** on GSM8K versus 76.7% for SFT. [Project] [Code]

Skills

Programming: Python, C++ — **ML/Robotics:** PyTorch, MuJoCo, Isaac Gym, Isaac Lab — **Systems:** Slurm, Docker, Linux